

Top 50 C Programming Problems

This list is designed to cover the **most important C concepts + basic problem-solving patterns** students should know before moving toward DSA and coding interviews.

● Fundamentals & Operators

1. Add, subtract, multiply and divide two numbers
2. Calculate area and perimeter of a rectangle
3. Calculate simple interest
4. Swap two numbers using a third variable
5. Swap two numbers without using a third variable
6. Convert Celsius to Fahrenheit and vice versa
7. Find the average of N numbers
8. Check whether a number is positive, negative or zero
9. Check whether a number is even or odd
10. Find the largest of three numbers

● Conditions & Decision Making

11. Check whether a year is a leap year
12. Check whether a character is a vowel or consonant
13. Check whether three sides form a valid triangle
14. Identify equilateral, isosceles or scalene triangle
15. Calculate student grade from marks
16. Create a simple calculator
17. Calculate electricity bill using slabs
18. Check whether a number is divisible by 5 and 11

● Loops & Number Logic

19. Print numbers from 1 to N
20. Print all even and odd numbers from 1 to N
21. Find the sum of numbers from 1 to N
22. Find the factorial of a number
23. Print the multiplication table of a number
24. Count the number of digits in a number
25. Find the sum of digits of a number
26. Find the product of digits of a number
27. Reverse a number
28. Check whether a number is a palindrome
29. Find the largest digit in a number

30. Count even, odd and zero digits in a number

Important Number Problems

- 31. Check whether a number is prime
- 32. Print all prime numbers between 1 and N
- 33. Find all factors of a number
- 34. Find the GCD/HCF of two numbers
- 35. Find the LCM of two numbers
- 36. Generate the Fibonacci series
- 37. Check whether a number is an Armstrong number
- 38. Check whether a number is a perfect number
- 39. Check whether a number is a strong number
- 40. Find the digital root of a number

Nested Loops & Patterns

- 41. Print a square pattern of stars
- 42. Print a right-angled triangle of stars
- 43. Print an inverted triangle of stars
- 44. Print a pyramid pattern
- 45. Print a number triangle
- 46. Print Floyd's Triangle
- 47. Print a 1-0 pattern
- 48. Print a multiplication table grid

Mini Projects / Logic Building

- 49. **Number Guessing Game** — random number + attempts + hints
- 50. **ATM / Restaurant Billing System** — menu, loops, conditions, calculations